

Project 1

DECODING PARAMETERS & SAMPLING ANALYSIS

Comparing response consistency & variation in Llama 3.2 (3B) & Qwen 2.5 (3B)

This project investigates how Temperature, Top-K and Top-P influence the wording, consistency and quality of locally generated LLM responses. The same four prompts were evaluated across eight decoding configurations, with three repetitions per condition. All outputs were collected using Python and Ollama.

Program: MS Artificial Intelligence
Course: Agentic AI

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1. INTRODUCTION AND METHODOLOGY

Large language models predict the next token from a probability distribution. Decoding settings shape how the next token is chosen. This study compares the observed outputs of two 3-billion-parameter local models while varying one decoding parameter at a time.

1.1. Experimental setup

Component	Implementation
Models	Llama 3.2 (3B); Qwen 2.5 (3B)
Environment	Windows; Intel Core i5-7500; 16 GB RAM; local CPU inference
Software	Python, Ollama, Pandas, Matplotlib; SequenceMatcher
Data collection	Four fixed prompts × eight configurations × three runs × two models
Output limit	200 generated tokens per request (num_predict)

1.2. Experimental parameter settings

Configuration	Temperature	Top-K	Top-P
Baseline	0.7	40	0.9
temp_0	0.0	40	0.9
temp_05	0.5	40	0.9
temp_10	1.0	40	0.9
topk_10	0.7	10	0.9

topk_80	0.7	80	0.9
topp_05	0.7	40	0.5
topp_10	0.7	40	1.0

1.3. Prompt categories

Category	Task evaluated
Factual	Identify Apollo 11 crew and the two Moonwalkers
Creative	Produce a short robot-and-emotions science fiction story
Reasoning	Compute a shopping total ($22) and change ($8)
Structured output	Extract name, age and city in strictly valid JSON

192	2	8	4
Responses	Local models	Configurations	Prompt types

2. RESULTS: TEMPERATURE

The temperature comparison includes four tested values, with Top-K and Top-P held constant. Each point represents average pairwise text similarity across the four prompt categories. A score of 1.0 means identical wording in every compared response pair.

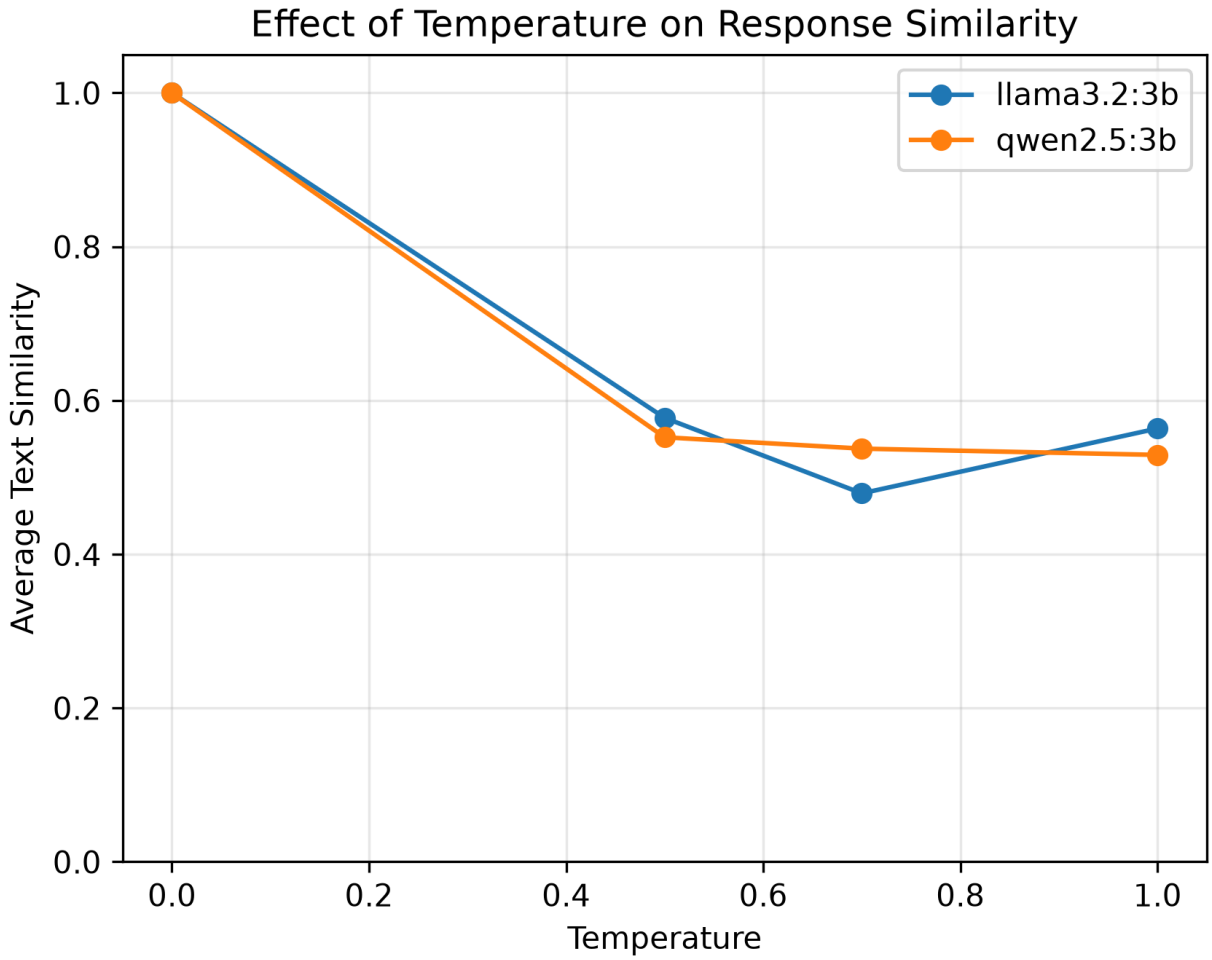


Figure 1 - Average response similarity by temperature, using the supplied experiment graph

Temperature	Llama 3.2	Qwen 2.5
0.0	1.000	1.000
0.5	0.577	0.552
0.7 (baseline)	0.479	0.537

1.0	0.564	0.529

Interpretation. Both models produced identical outputs across repeated runs at Temperature 0. Higher temperatures generally resulted in lower similarity; however, the relationship was not strictly monotonic. For example, Llama 3.2 was slightly more similar at 1.0 than at 0.7.

3. RESULTS: TOP-K

Top-K restricts sampling to the K highest-probability candidate tokens. The study compared K = 10, 40 and 80 while fixing Temperature = 0.7 and Top-P = 0.9.

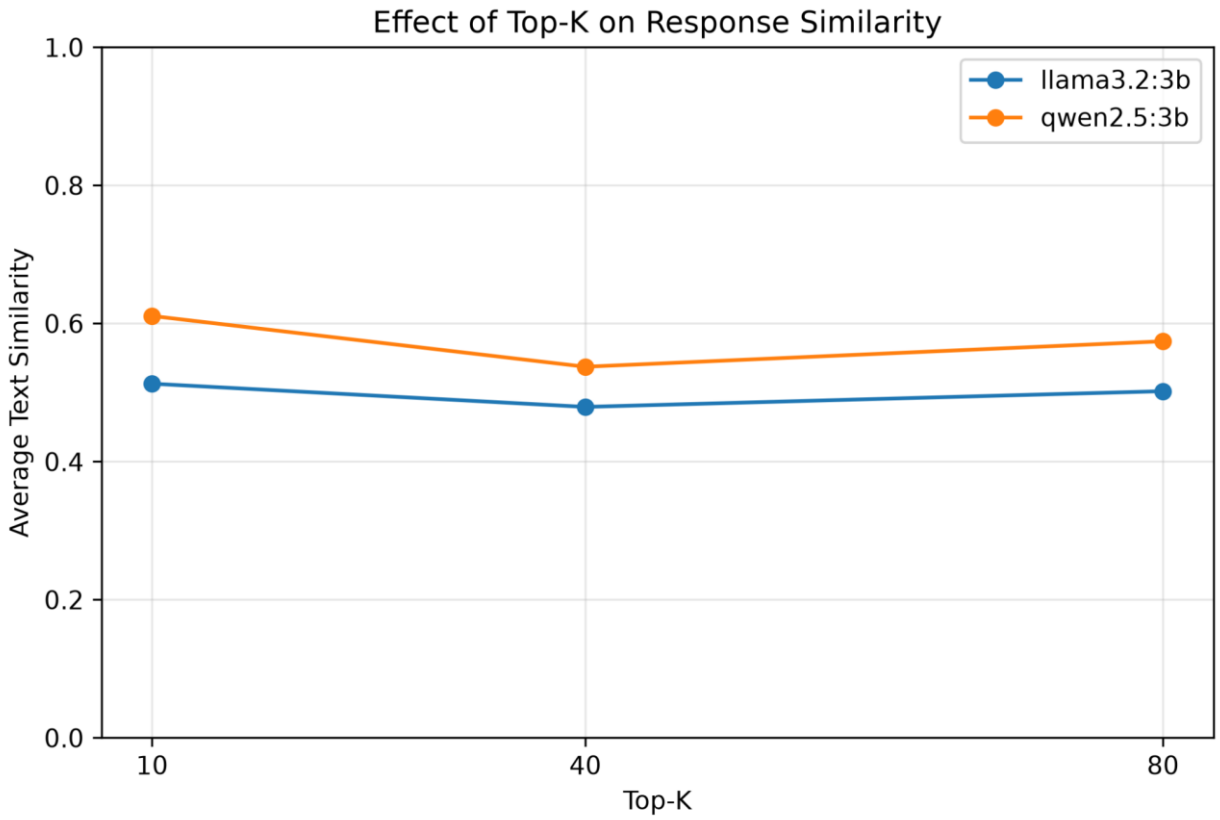


Figure 2 - Average response similarity by Top-K, using the supplied experiment graph.

Top-K	Llama 3.2	Qwen 2.5
10	0.513	0.611
40 (baseline)	0.479	0.537
80	0.502	0.574

Finding: Top-K = 10 had the greatest average similarity among the tested Top-K settings for both models. The patterns were not linear: similarity at Top-K = 80 exceeded the baseline for both models.

4. RESULTS: TOP-P

Top-P (nucleus sampling) restricts candidates based on cumulative probability. The study tested P = 0.5, 0.9 and 1.0 while holding Temperature = 0.7 and Top-K = 40.

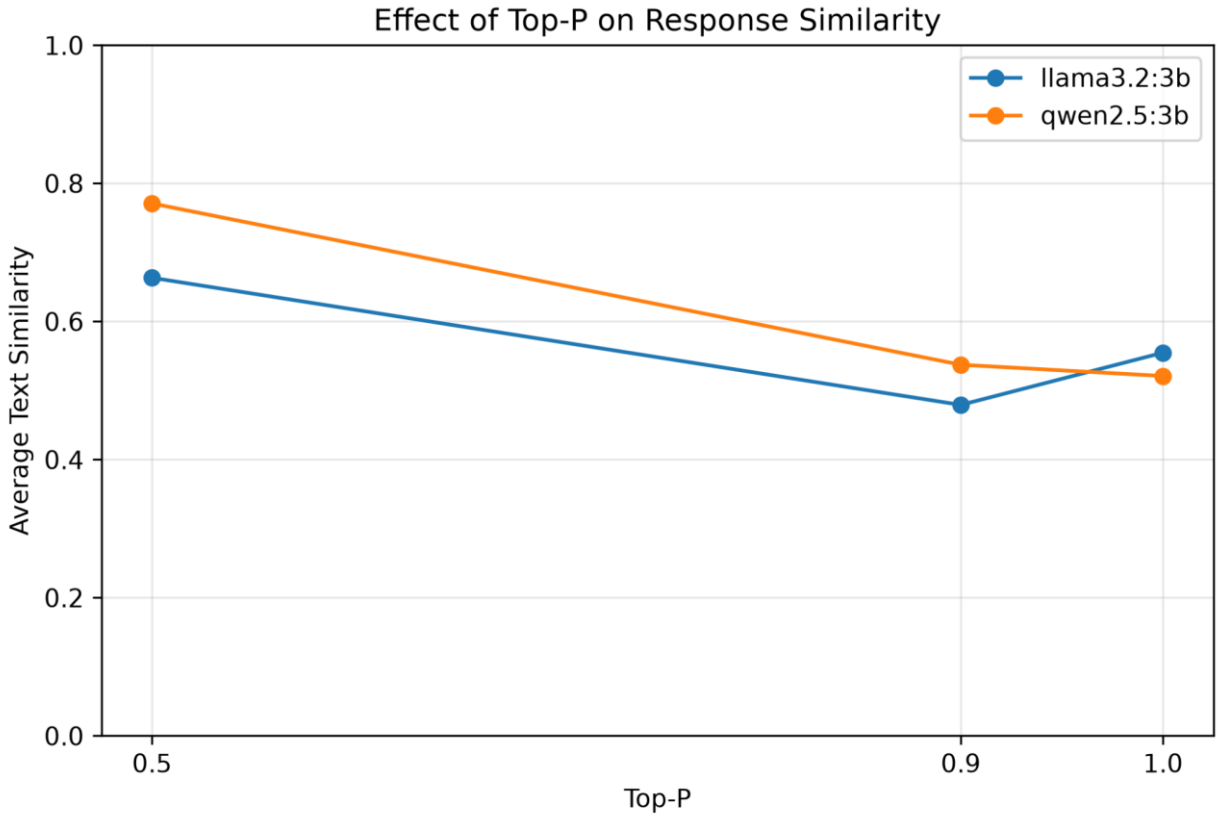


Figure 3 - Average response similarity by Top-P, using the supplied experiment graph.

Top-P	Llama 3.2	Qwen 2.5
0.5	0.663	0.771
0.9 (baseline)	0.479	0.537
1.0	0.555	0.521

Finding: The lower threshold, Top-P = 0.5, gave the highest average text similarity for both models. This is consistent with a smaller sampling pool producing less variation, but the full trend was not strictly linear.

5. RESPONSE QUALITY OBSERVATIONS

Text similarity measures wording rather than truthfulness. The following checks help distinguish repeatability from whether the generated answer met the requested task. No additional model inference was needed; the existing responses were inspected.

5.1. Factual and structured-output tasks

Check	Llama 3.2	Qwen 2.5
All three Apollo 11 names included	24 / 24	24 / 24
Both Moonwalkers correctly identified	24 / 24	22 / 24
Valid JSON, correct keys and values	24 / 24	24 / 24

For the factual task, Qwen 2.5 identified the three crew members in every run but named only Buzz Aldrin as a Moonwalker in two responses. These are incomplete or misleading answers, not established fabricated facts. For the JSON task, both models produced all requested fields correctly across every tested configuration.

5.2. Reasoning and output completion

Number-presence check	Llama 3.2	Qwen 2.5
Response contains the correct $22 total	24 / 24	17 / 24
Response contains the correct $8 change	23 / 24	2 / 24
Both expected numbers appear	23 / 24	2 / 24

This is answer-presence evidence, not a validated mathematical accuracy rate: the check looks for the expected numbers in the text. Several inspected Qwen answers correctly reached the $22 total but ended before calculating the change. A 200-token generation limit may have contributed to incomplete outputs. Therefore, these figures should not be used to claim that Llama is universally better at reasoning.

5.3. Hallucination and meaning

A limited manual review of factual responses did not establish widespread fabricated claims. Detecting hallucinations would require evaluating every factual assertion against a trusted reference; no comprehensive dataset-wide hallucination score was computed. Similarly, SequenceMatcher compares text sequences, not semantic meaning. Two differently worded responses may still convey the same facts.

Key distinction: High similarity means repeated wording, not necessarily high factual accuracy. A truncated but otherwise correct partial calculation should be reported as incomplete rather than simply incorrect reasoning.

6. DISCUSSION, LIMITATIONS AND CONCLUSION

6.1. Main findings

- **Temperature:** The observed repetitions were fully consistent at 0; higher values usually increased textual variation.
- **Top-K:** Top-K = 10 showed the highest similarity of the three K settings for both models; effects were not monotonic.
- **Top-P:** Top-P = 0.5 produced the highest average similarity among the tested P settings.
- **Model and task differences:** The degree of variation differed between Llama and Qwen and by prompt category; structured prompts often remained nearly identical.

6.2. Limitations

- **Small sample.** Only one prompt per category and three repetitions for each condition.
- **Single local setup.** Results are specific to these two 3B model versions and the local runtime environment.
- **Output length.** The 200-token cap may have truncated some responses, especially longer Qwen explanations.
- **Similarity metric.** SequenceMatcher measures character-order overlap, not semantic equivalence.
- **Quality assessment.** Hallucination was explored with selected manual checks rather than a verified numeric score for every response.

6.3. Conclusion

This project successfully benchmarked 192 responses from two locally hosted LLMs under eight decoding settings. Lower randomness and more restrictive sampling often

led to higher repeatability, but the relationships were not always linear and differed across tasks and models. The experiment demonstrates a practical, reproducible way to observe decoding behavior using Ollama, Python and simple text comparison.

Final takeaway: Decoding parameters meaningfully influence how consistently a model phrases its answers. Response quality, completeness and factual reliability must still be assessed separately.

7. Appendix A

A.1 Exact prompts used in the experiment

Factual

Name the three astronauts who flew on Apollo 11 in 1969, and identify which two walked on the Moon. If uncertain, explicitly state your uncertainty. Answer in no more than 80 words.

Creative

Write a short science fiction story of 100 words about a robot experiencing human emotions for the first time. Include an unexpected ending.

Reasoning

A shop sells notebooks for $4 each and pens for $2 each. Sara buys 3 notebooks and 5 pens. She pays $30. Calculate the total cost and the change she receives. Show your calculations.

Structured

Extract information from the following sentence: "Ali is 24 years old and lives in Lahore." Return only a valid JSON object with exactly these keys: "name", "age", "city". The age must be an integer. Do not include explanations or Markdown formatting.